

GLMs for Applied Research

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Session 7: Multinomial & Ordinal Logistic Regression

Topics we will cover in this session

- Nominal vs ordinal outcomes
- Multinomial logit model and baseline categories
- Proportional odds model for ordinal data
- Fitting in R, diagnostics check, and interpreting the models

Why we need more than binary logistic regression

Binary logistic regression

- Outcome Y has exactly two categories (e.g., yes / no)
- Models the log-odds of the outcome as a linear function of predictors

$$\log[P(Y=1) / P(Y=0)] = \beta_0 + \beta_1 X_1 + \dots + \beta_k X_k$$

- Coefficients exponentiate to odds ratios

But many outcomes have 3+ categories

N **Nominal outcome — categories have no order**

*Commute mode: Car, Bus, Bike, Walk →
Multinomial Logistic Regression*

O **Ordinal outcome — categories have a natural order**

*Satisfaction: Low, Medium, High → Ordinal
Logistic Regression*

Nominal vs ordinal outcomes

N Nominal (unordered)

- Transportation mode: Car / Bus / Bike / Walk
- Brand choice: Brand A / Brand B / Brand C
- Blood type: A / B / AB / O
- Marital status: Single / Married / Divorced / Widowed

O Ordinal (ordered)

- Customer satisfaction: Low / Medium / High
- Education level: HS / Bachelor's / Graduate
- Disease stage: Stage I / II / III / IV
- Likert item: Disagree / Neutral / Agree

The baseline-category logit model

With J outcome categories, choose one as the reference (baseline) category — usually the most common or most substantively meaningful group. The model fits $J-1$ logit equations simultaneously, each comparing one category to the baseline:

$$\log[P(Y=j) / P(Y=J)] = \beta_{0j} + \beta_{1j} X_1 + \beta_{2j} X_2 + \dots + \beta_{kj} X_k$$

for $j = 1, \dots, J-1$ — fit simultaneously for every non-reference category

Category-specific coefficients

Each non-reference category j gets its own set of β 's, estimated jointly with all others.

J is the baseline

All comparisons are made relative to whichever category you set as reference.

Simultaneous estimation

Fit by maximum likelihood as one joint model, not $J-1$ separate binary regressions.

Interpreting coefficients as relative risk

$\exp(\beta_{kj})$ = Relative Risk Ratio (RRR)

the multiplicative change in the odds of category j vs. baseline J , per one-unit increase in X_k

- $RRR > 1 \rightarrow$ a one-unit increase in X_k makes category j more likely relative to the baseline
- $RRR < 1 \rightarrow$ a one-unit increase in X_k makes category j less likely relative to the baseline
- RRR is always relative to whichever baseline you chose — re-leveling the outcome changes every RRR
- There is no single “overall” effect of X_k — effects must be read one non-reference category at a time

Fitting the model in R

Commute-mode example: Car / Bus / Bike, predicted by income, age, and distance to work (baseline = "Car")

```
library(nnet)
df$mode <- relevel(df$mode, ref = "Car")
model_mn <- multinom(mode ~ income + age + distance, data = df)
summary(model_mn)      # coefficients + std. errors
exp(coef(model_mn))    # relative risk ratios
```

relevel() sets the baseline

Choose the reference category explicitly before fitting — nnet uses the first factor level by default.

multinom()

Fits a single-layer neural net with no hidden units, which is mathematically equivalent to the multinomial logit MLE.

No p-values by default

summary() gives coefficients & SEs only; compute $z = \text{coef}/\text{SE}$ and compare to the normal distribution.

Reading the output

Relative risk ratios, each vs. baseline category “Car”:

Predictor	RRR: Bus vs. Car	RRR: Bike vs. Car	Interpretation
Income (\$1k)	0.85	0.60	Higher income → lower odds of Bus or Bike vs. Car
Age (years)	0.97	0.93	Older commuters favor Car relative to both alternatives
Distance (mi)	1.12	0.81	Longer distance → more Bus, less Bike, relative to Car

READING TIP

An RRR of 0.60 for income on “Bike vs. Car” means: for each additional \$1,000 of income, the odds of choosing Bike over Car are multiplied by 0.60 (a 40% decrease) — holding age and distance constant. This says nothing directly about Bike vs. Bus; that comparison requires re-leveling the baseline or computing it from the two available equations.

The IIA assumption and model diagnostic

Independence of Irrelevant Alternatives (IIA)

The relative odds between any two categories are assumed not to depend on what other alternatives are available.

Classic illustration — the “red bus / blue bus” paradox: adding a near-identical alternative shouldn't change the odds between existing categories, but intuitively it does.

Checking Model Fit

- Hausman-McFadden test — formally tests the IIA assumption by comparing full vs. restricted-choice-set models
- Likelihood ratio test — compares the fitted model to an intercept-only null model
- McFadden's pseudo- R^2 — $1 - (\log\text{-likelihood full} / \log\text{-likelihood null})$; values of 0.2–0.4 indicate good fit
- AIC / BIC — for comparing non-nested candidate models

The proportional odds (cumulative logit) model

For an ordinal outcome with K ordered categories, the model works with cumulative probabilities — the probability of being at or below category k — rather than one category at a time:

$$\text{logit}[P(Y \leq k)] = \alpha_k - (\beta_1 X_1 + \beta_2 X_2 + \dots + \beta_p X_p)$$

for $k = 1, \dots, K-1$ — same slopes β , a different threshold α_k at each cutpoint

One set of slopes

Unlike the multinomial model, the β 's do not vary by cutpoint k — only the intercepts α_k do.

$K-1$ intercepts

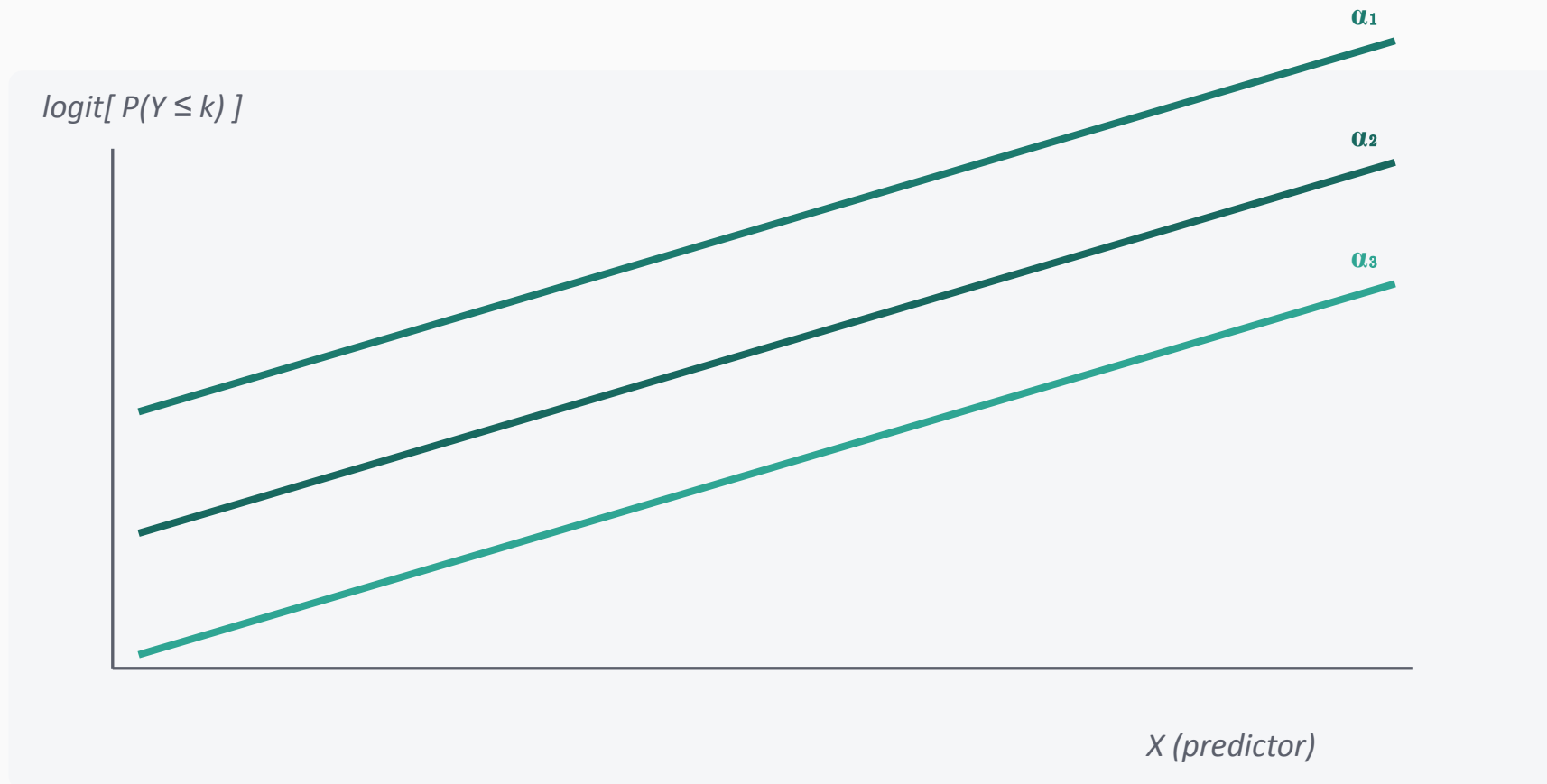
Each cutpoint between adjacent categories gets its own threshold α_k , ordered $\alpha_1 < \alpha_2 < \dots$

“Proportional odds”

The name reflects that the odds ratio for a predictor is the same across every cutpoint in the scale.

Visualizing the proportional odds assumptions

Each cutpoint defines its own S-shaped probability curve. Proportional odds means every curve has the same shape and slope — just shifted left or right by its own intercept α_k :



If the lines aren't parallel

The effect of a predictor differs across cutpoints — the proportional odds assumption is violated, and a single odds ratio per predictor is misleading.

Three parallel lines, one per cutpoint ($\alpha_1, \alpha_2, \alpha_3$) — identical slope β , shifted thresholds

Fitting the model in R

Satisfaction example: Low / Medium / High, predicted by income, age, and service quality rating

```
library(MASS)
model_or <- polr(satisfaction ~ income + age + service_quality,
                 data = df, Hess = TRUE)
summary(model_or)      # coefficients + thresholds (intercepts)
exp(coef(model_or))    # odds ratios
exp(confint(model_or)) # 95% CI for odds ratios
```

satisfaction must be an ordered factor

Use `factor(..., ordered = TRUE)` with levels in the correct order before fitting.

Hess = TRUE

Returns the observed Hessian, needed to compute standard errors for `summary()` and `confint()`.

method =

Defaults to `logit`; set `method="probit"` or `"cloglog"` for alternative link functions.

Interpreting the output

$\exp(\beta_k) = 1.42$ for service_quality

→ a one-unit increase in service quality is associated with 42% higher odds of being in a higher (or equal) satisfaction category — the same 1.42 applies at every cutpoint

- Because of proportional odds, one odds ratio per predictor summarizes its effect across all category boundaries
- Thresholds α_k (the “intercepts”) mark the boundaries between adjacent categories on the latent logit scale — not usually of direct interest
- `polr()` does not print p-values by default: compute $t = \text{coefficient} / \text{std. error}$ and compare to a standard normal, or refit with p-values via `pt()`
- Model fit: compare nested models with `anova()`, or use AIC / pseudo- R^2 for non-nested comparisons

Testing the proportional odds assumption

The proportional odds assumption is testable — always check it before trusting a single set of odds ratios.

```
library(brant)
brant(model_or)  #  $H_0$ : proportional odds holds for every predictor
```

IF THE ASSUMPTION HOLDS

Brant test $p > .05$ for all predictors (and the omnibus test) → a single proportional-odds model is appropriate. Report one odds ratio per predictor with confidence.

IF THE ASSUMPTION IS VIOLATED

Consider a partial proportional odds model (`vglm()` in package VGAM, allowing select predictors to vary by cutpoint), or fall back to multinomial logistic regression, treating the outcome as unordered.

Choosing the right model for your outcome

	Multinomial Logit	Ordinal Logit (Proportional Odds)
Outcome structure	Unordered categories	Ordered categories
Coefficients	One set per non-reference category	One shared set + K-1 thresholds
Key assumption	IIA	Proportional odds
R function	<code>nnet::multinom()</code>	<code>MASS::polr()</code>
Effect size	Relative risk ratio	Odds ratio (constant across cutpoints)

DECISION RULE

Is the outcome ordered? If no → multinomial logit. If yes → fit ordinal logit, then run `brant()`; if proportional odds holds, keep it — if not, use a partial proportional odds model or fall back to multinomial.

Recap & What's next

- Nominal outcomes (no order) → multinomial logit; ordinal outcomes (ordered) → proportional odds model
- Multinomial: category-specific coefficients, interpreted as relative risk ratios vs. a chosen baseline; watch the IIA assumption
- Ordinal: one shared coefficient per predictor plus $K-1$ thresholds, interpreted as a single odds ratio; always test proportional odds with `brant()`
- When the proportional odds assumption fails, consider partial proportional odds models or treat the outcome as multinomial

Next: Hands-on 7 Notebook, practical exercise.